

SIMATS ENGINEERING

ASSESSMENT TOOL 3: Comparative Analysis

COURSE NAME: COMPUTER VISION COURSE

CODE: ITA0509

COURSE OUTCOME COVERED

CO4: Compare object and pattern recognition techniques for interpreting visual data with spatial and motion characteristics. (BTL 5)

Assessment Tool Weightage: 10%

QUESTIONS: 5 Total Marks:50

Annexure A Comparative Analysis

Code of Conduct:

I N.AKSHAYA(192425129) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Q. No	Scope of Items (20%)	Comparison Depth (25%)	Use of Evidence (20%)	Critical Thinking (20%)	Clarity of Presentation (15%)	Total Score (100%)
1						
2						
3						
4						
5						
Total Marks (Out of 100)				Total Marks (Out of 50)		

Annexure A

Comparative Analysis

Compute the required performance measures from the given data and compare the techniques based on the results. Evaluate and justify your conclusion using the calculated values.

1. Detection Performance Comparison

Two object detection models produce the following results:

Model True Positives False Positives False Negatives

A (Viola-Jones) 80 20 30

B (YOLO) 90 15 20

(a) Compute precision and recall for both models.

(b) Compare their performance and state which model is better for object detection.

Answer:

Given:

Model	TP	FP	FN
Viola-Jones (A)	80	20	30
YOLO (B)	90	15	20

(a) Calculate Precision and Recall

Formula for Precision:

$$\text{Precision} = \text{TP} / (\text{TP} + \text{FP})$$

Formula for Recall:

$$\text{Recall} = \text{TP} / (\text{TP} + \text{FN})$$

Model A – Viola-Jones

Precision:

$$\text{Precision} = 80 / (80 + 20)$$

$$= 80 / 100$$

$$= \mathbf{0.80 = 80.00\%}$$

Recall:

$$\text{Recall} = 80 / (80 + 30)$$

$$= 80 / 110$$

$$= \mathbf{0.7273 = 72.73\%}$$

Model B – YOLO

Precision:

$$\text{Precision} = 90 / (90 + 15)$$

$$= 90 / 105$$

$$= \mathbf{0.8571 = 85.71\%}$$

Recall:

$$\text{Recall} = 90 / (90 + 20)$$

$$= 90 / 110$$

$$= \mathbf{0.8182 = 81.82\%}$$

(b) Comparison

Measure	Model A: Viola-Jones	Model B: YOLO	Better
Precision	80.00%	85.71%	YOLO
Recall	72.73%	81.82%	YOLO

Conclusion: YOLO performs better for object detection because it has both **higher precision and higher recall**. Its higher precision means it produces fewer false positives, while its higher recall means it detects a greater proportion of the actual objects.

2. Motion Estimation Error Analysis

Two motion estimation techniques produce displacement errors (in pixels):

Frame Method A Method B

1 2 3

2 3 5

3 2 4

(a) Compute average error for both methods.

(b) Compare and evaluate which method is more accurate.

Answer:

The given displacement errors are:

Frame	Method A	Method B
1	2	3
2	3	5
3	2	4

(a) Calculate Average Error

Formula:

Average Error = Sum of Errors / Number of Frames

Method A

Average Error = $(2 + 3 + 2) / 3$

= $7 / 3$

= **2.33 pixels**

Method B

Average Error = $(3 + 5 + 4) / 3$

= $12 / 3$

= **4.00 pixels**

(b) Comparison

Method	Average Error
Method A	2.33 pixels
Method B	4.00 pixels

Conclusion: Method A is more accurate because it has a lower average displacement error. Its average error is 2.33 pixels compared with 4.00 pixels for Method B. Therefore, Method A provides approximately **1.67 pixels less error on average**.

3. Optical Flow Magnitude Comparison

Optical flow magnitudes (in pixels/frame):

Region Method A Method B

R1 5 8

R2 6 10

R3 5 9

(a) Compute average motion magnitude for both methods.

(b) Compare and evaluate which method captures stronger motion.

Answer:

The given optical-flow magnitudes are:

Region	Method A	Method B
R1	5	8
R2	6	10
R3	5	9

(a) Calculate Average Motion Magnitude

Formula:

Average Motion Magnitude = Sum of Magnitudes / Number of Regions

Method A

Average = $(5 + 6 + 5) / 3$

= $16 / 3$

= **5.33 pixels/frame**

Method B

Average = $(8 + 10 + 9) / 3$

= $27 / 3$

= **9.00 pixels/frame**

(b) Comparison

Method	Average Motion Magnitude
Method A	5.33 pixels/frame
Method B	9.00 pixels/frame

Conclusion: Method B captures stronger motion because its average optical-flow magnitude is 9.00 pixels/frame, compared with 5.33 pixels/frame for Method A.

The difference is:

Therefore, Method B indicates substantially stronger estimated motion across the three regions.

Important interpretation: A higher optical-flow magnitude means stronger estimated motion; it does **not automatically mean greater accuracy**. The question specifically asks which method captures stronger motion.

4. Pattern Recognition Accuracy Comparison

Two recognition systems produce results:

System Correct Predictions Total Samples

Traditional 85 100

Deep Learning 92 100

(a) Compute accuracy for both systems.

(b) Compare and evaluate which system performs better.

Answer:

The given results are:

System	Correct Predictions	Total Samples
Traditional	85	100
Deep Learning	92	100

(a) Calculate Accuracy

Formula:

$$\text{Accuracy} = (\text{Correct Predictions} / \text{Total Samples}) \times 100$$

Traditional System

$$\text{Accuracy} = (85 / 100) \times 100$$

$$= 85\%$$

Deep Learning System

$$\text{Accuracy} = (92 / 100) \times 100$$

$$= 92\%$$

(b) Comparison

System	Accuracy
Traditional	85%
Deep Learning	92%

Conclusion: The **Deep Learning system performs better**, achieving 92% accuracy compared with 85% for the Traditional system.

The improvement is:

Thus, based on the provided dataset, Deep Learning provides more correct pattern-recognition predictions.

5. Detection Accuracy Comparison

Results:

Model Correct Total

A 180 200

B 170 200

(a) Compute accuracy.

(b) Compare and evaluate.

Answer:

Given results:

Model	Correct	Total
A	180	200
B	170	200

(a) Calculate Accuracy

Formula:

$$\text{Accuracy} = (\text{Correct Predictions} / \text{Total}) \times 100$$

Model A

$$\text{Accuracy} = (180 / 200) \times 100$$

$$= 90\%$$

Model B

$$\text{Accuracy} = (170 / 200) \times 100$$

$$= 85\%$$

(b) Comparison

Model	Accuracy
Model A	90%
Model B	85%

Conclusion: Model A performs better because its accuracy is 90%, which is 5 percentage points higher than Model B's 85%.

Therefore, based solely on the provided accuracy results, Model A is the preferred detection model.

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Scope of Items	20%	Selects highly relevant items with clear scope and justification	Selects appropriate items with minor gaps	Selection is limited or partially relevant	Items are unclear or not relevant
Comparison on Depth	25%	Provides detailed and comprehensive comparison across multiple aspects	Provides adequate comparison with some depth	Limited comparison with few aspects	Very minimal or no comparison
Use of Evidence	20%	Supports comparison with accurate data, examples, or references	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Critical Thinking	20%	Demonstrates strong critical thinking with clear interpretation and reasoning	Shows reasonable interpretation with minor gaps	Basic interpretation ; lacks depth	No meaningful interpretation
Clarity of Presentation	15%	Well organized, clear, and logically structured presentation	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand